

Digital Conductivity Sensor

Basic User Manual



Model: DEC890

Version 1.0

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Chapter 1 Specification

Product specifications are subject to change without notice.

Measuring principle	Graphite two-electrode measurement principle
Measuring range	0~20uS/cm, 20~200uS/cm, 200~2000uS/cm, 2~10mS/cm 0.0~50.0℃
Resolution	0.01uS/cm, 0.1uS/cm, 1uS/cm 0.1℃
Accuracy	±1% of the measuring range
TEMP compensation	0~50℃, automatic or manual compensation
Calibration method	Offset calibration, factor calibration, multi-point calibration
Operating temperature	0 to 50℃
Work pressure	0~1Bar
Waterproof level	IP68
Power requirements	DC 9~36V
Power consumption	About 0.2W
Electrical isolation	Power supply and communication are isolated
Communication Interface	RS485 MODBUS
Shell material	Stainless steel 316, POM
Case size	Diameter 37mm, total length about 261mm
Installation size	1-inch NPT thread at both ends
Cable	Standard 10 meters, length can be customized
Connector	M16

Chapter 2 Basic Information

2.1 Security Information

Please read this manual completely before unpacking, installing and operating this equipment. Pay special attention to all precautions. Otherwise, it may cause serious personal injury to the operator or damage the equipment.

2.2 Overview

This sensor is based on the two-electrode measurement principle. Two parallel plates are placed into the solution to be measured, a driving voltage is applied to both ends of the plates, and the conductivity value of the solution is measured based on the current intensity between the plates. The sensor electrode plate is made of graphite material, which can minimize the polarization effect and ensure measurement accuracy. The sensor measurement unit is designed as a replacement structure to reduce the replacement and maintenance costs during user use.

The sensor is suitable for sewage treatment plants, water plants, water stations, surface water, environmental water treatment, metallurgy electronics, mining, paper industry, semiconductors and other fields, and can continuously monitor the conductivity of aqueous solutions.

2.3 Dimensions

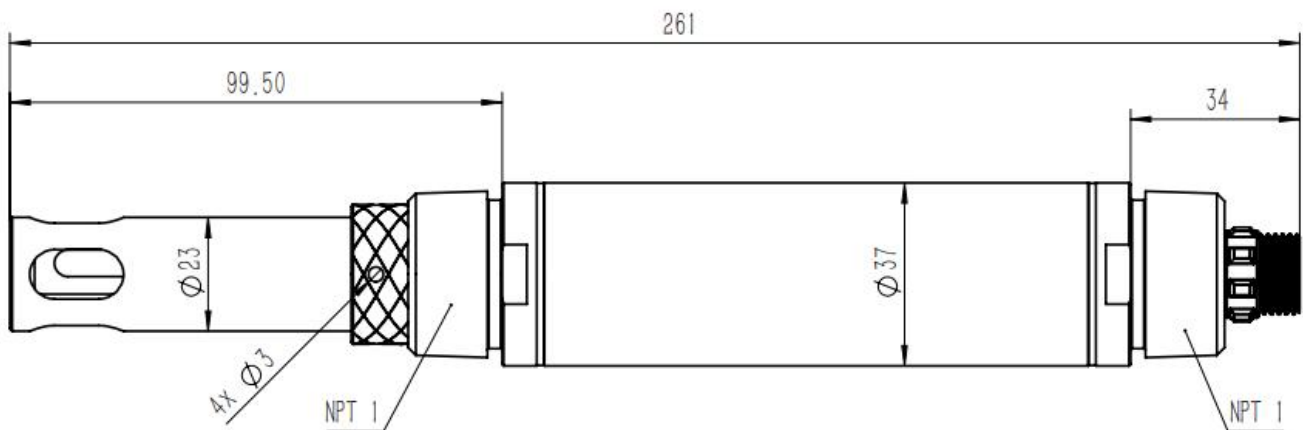


Figure 1 Dimensions of the sensor

Chapter 3 Installation

3.1 Sensor Installation

Refer to the pictures in this section to install and fix the sensor. To ensure that the sensor can measure safely and accurately, please meet the following conditions during installation:

- When installing in pipeline, the sensor should be installed at a lower position in the pipeline, where the flow rate is stable and bubbles are not easily generated.
- When installing in submerged mode, the front section of the sensor should be installed facing the direction of the water flow and should be deep into the moving water body.
- When installing, the sensor measuring part must be kept clean. Do not touch the inner surface of the sensor with your hands or unclear objects. Contact with oily or glue-like objects will result in inaccurate measurement for a long time.
- The sensor wire is prohibited from being piped and tied together with the power line to avoid interference with the measurement or breakdown of the sensor measurement unit when damaged.

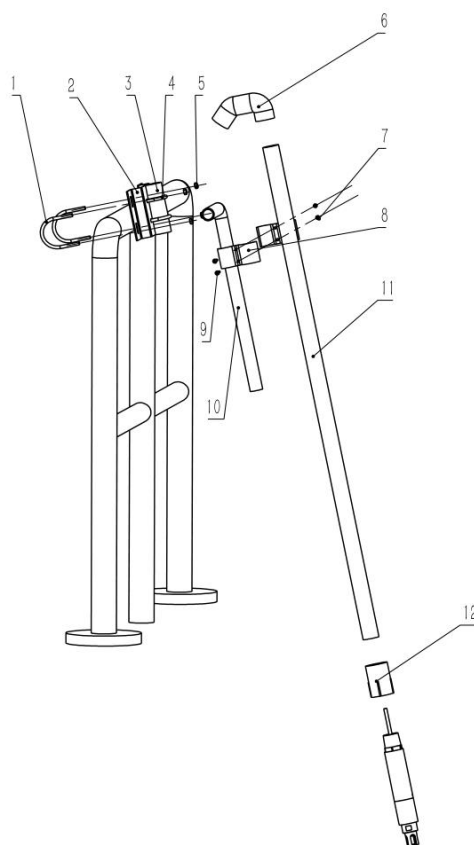


Figure 2 Schematic diagram of railing installation

1- DN60 U-shaped card	7- M4 screw nut*2
2- “┐”shaped board	8- “8”shaped clip 25&32
3- Handle sleeve	9- M4*25 screw*2
4- DN40 U-shaped card	10- Handle
5- M6 screw nut*8	11- DN32 PVC Bracket
6- Rainproof elbow	12-1 inch inner wire straight pipe joint

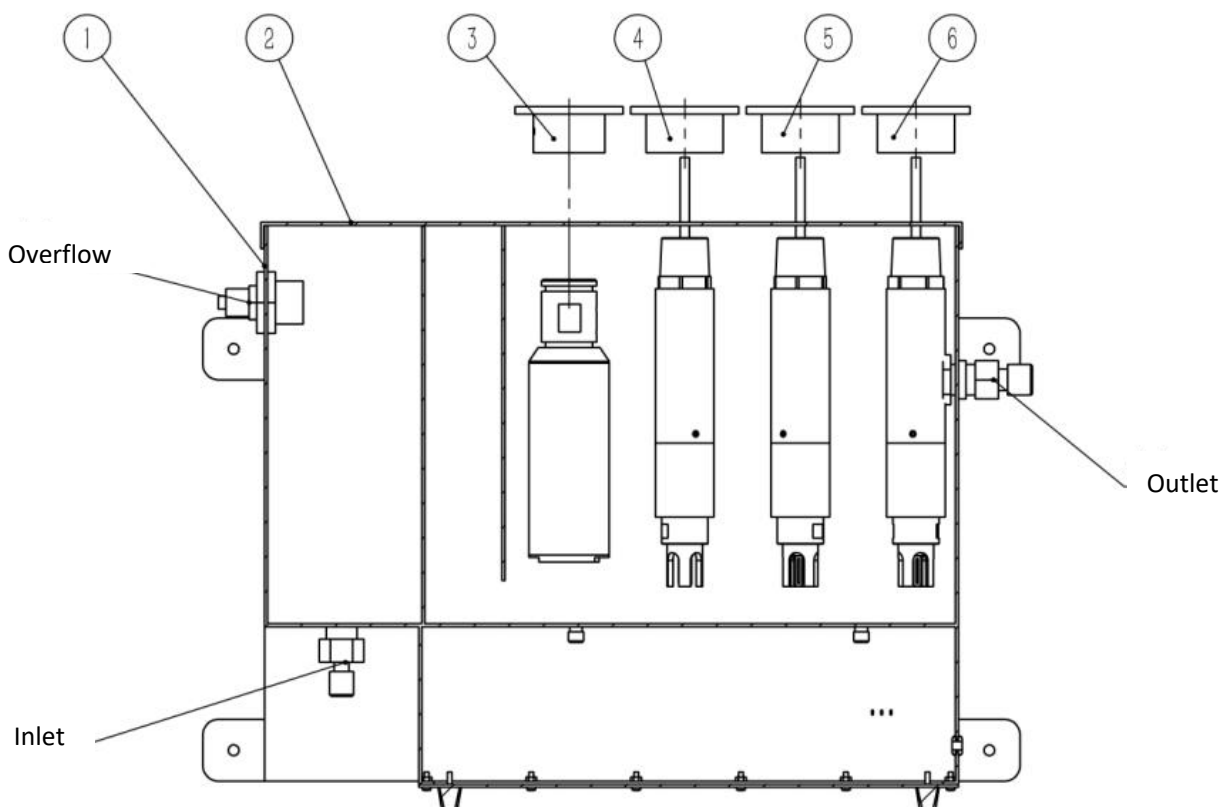


Figure 3 Schematic diagram of flow cell installation

1-Flow cell	4-pH sensor fixed connection cover
2-Flow cell cover	5-DO sensor fixed connection cover
3-Turbidity sensor fixed connection cover	6-Conductivity sensor fixed connection cover

3.2 Sensor Wiring

The sensor is correctly connected as defined in the table below.

Note: Aviation plug version does not require user to connect wires.

Wire Color	Red	Black	White	Green
Terminal Definition	Power positive	Power negative	RS485 data A (+)	RS485 data B (-)
Terminal Symbols	V+	V-	AS	BS

Chapter 4 Use

4.1 Communication Protocol

The sensor communication is RS485 Modbus-RTU. For the specific communication protocol, please refer to the Modbus related protocol description. The default communication parameters of the sensor are: communication address=17, baud rate=9600, check bit=none, stop bit=1. The communication parameters can be modified by referring to the register description in appendix A.

4.2 Read Measured Values

The sensor measurement value can be read by connecting the controller or using other Modbus master devices. For the Modbus register address, please refer to Appendix A. The sensor measurement value data is 4-byte floating point data, and the data order is little endian-byte exchange. Pay attention to the conversion order.

4.3 Sensor Calibration

The sensor provides two measurement and calibration methods, two-point calibration and multi-point calibration. Normally, two-point calibration can meet the daily calibration and maintenance needs. Multi-point calibration is required when the sensor measurement deviation is large or the measurement unit is replaced.

4.3.1 Two-points Calibration

Two-point calibration corrects the sensor measurement value through “offset value” and “factor value”. The relationship between the sensor measurement value and the two parameters is as follows:

Conductivity measurement value=(original measurement value+offset value)*factor value

Two-point calibration requires two water samples with known conductivity values. The calibration process is as follows:

- Use function code 06 to write value 33 to register 33 to restore calibration;
- Put the sensor in water sample 1 and gently shake the sensor to ensure that there are no bubbles at the sensor measurement end to affect the measurement. After the value stabilizes, record measurement value 1;
- Rinse the sensor with pure water and use filter paper to absorb the residual water. Put the sensor in water sample 2 and gently shake the sensor. After the value stabilizes, record measurement value 2;
- Calculate offset value and factor value

Factor value=(water sample value 2-water sample value 1)/(measurement value 2-measurement value 1)

Offset value=(water sample value 1/factor value)-measurement value 1

- Use function code 16 to write offset value and factor value to the corresponding register.

4.3.2 Multi-point Calibration

Multi-point calibration requires the preparation of four standard solutions: 84 μ S, 146 μ S, 1413 μ S, and 12880 μ S. The calibration process is as follows:

- Use deionized water to clean the sensor and absorb the residual water with filter paper. Place the sensor in dry air, wait for the measured value to stabilize, use the 16 function code, and write the value 0 to register 23.
- Use deionized water to clean the sensor and absorb the residual water with filter paper. Place the sensor in 84 μ S/cm conductivity standard solution, wait for the sensor measurement to stabilize, use the 16 function code, and write the value 84 to register 25.
- Use deionized water to clean the sensor and absorb the residual water with filter paper. Place the sensor in 146 μ S/cm conductivity standard solution, wait for the sensor measurement to stabilize, use the 16 function code, and write the value 146 to register 27.
- Use deionized water to clean the sensor and absorb the residual water with filter paper. Place the sensor in 1413 μ S/cm conductivity standard solution, wait for the sensor measurement to stabilize, use the 16 function code, and write the value 1413 to register 29.
- Use deionized water to clean the sensor and absorb the residual water with filter paper. Place the sensor in 12880 μ S/cm conductivity standard solution, wait for the sensor measurement to stabilize, use the 16 function code, and write the value 12880 to register 31.

Note: When calibrating, it is recommended to calibrate at 25°C, otherwise a small amount of temperature compensation error will be introduced.

Tips: The sensor supports up to 5-point calibration, and users can select calibration points according to the application.

4.3.3 Temperature Correction

Temperature correction provides the function of adjusting temperature deviation. The temperature deviation value is calculated as follows:

Temperature deviation value = actual temperature value - measured temperature value

4.3.4 Reset Calibration

Use function code 06 to write the value 33 to register 33 to restore the sensor calibration parameters to the factory state.

Chapter 5 Maintenance

The sensor contains precision measuring parts. Please ensure that the sensor is not subjected to any strong mechanical impact during use. There are no user-maintained parts inside the sensor.

5.1 Maintenance Cycle

Maintenance work	Maintenance frequency
Visual inspection	Monthly
Calibration check	Monthly(depending on the environmental conditions)
Seal maintenance	Every 2 years

5. 2 Clean

For routine cleaning, the sensor can be placed in clean water and shaken for cleaning. If the surface of the sensor measuring part is seriously contaminated, the sensor measuring part can be soaked in a light detergent or weak acid for 15 minutes, and then the electrode can be cleaned with distilled water.

Note: During installation and cleaning, do not use hard objects to touch the surface of the measuring electrode, and do not wipe the surface of the electrode with any object, otherwise it will cause inaccurate measurement and require re-calibration. In serious cases, it will affect the measurement range of the sensor.

Appendix A Modbus Register Information

Name	Register	Data type	Length	Access type	Function code	Description
Status	0	Unsigned integer	1	Read-only	04	When it is not 0, it means the sensor is being calibrated.
Conductivity measurement	2	floating point	2	Read-only	04	Output value unit uS/cm
Temp. measurement	4	floating point	2	Read-only	04	Temp. measurement, unit: °C
Salinity value	6	floating point	2	Read-only	04	Calculated value, unit: ppt
TDS value	8	floating point	2	Read-only	04	Calculated value, unit: ppm
Serial number	10	String	6	Read-only	04	12-bit ASCII serial number
Firmware version	16	Unsigned integer	1	Read-only	04	Version number x100
Address	0	Unsigned integer	1	Read/write	03/06	Communication address range 1~254, default 17
Baud rate	1	Unsigned integer	1	Read/write	03/06	0=4800,1=9600 (default), 2=19200,3=38400
Parity bit	2	Unsigned integer	1	Read/write	03/06	0=none (default), 1=odd, 2=even
Stop bit ⁽¹⁾	3	Unsigned integer	1	Read/write	03/06	1=1STOP (default), 2=2STOP
Average times	4	Unsigned integer	1	Read/write	03/06	0= 2 times (default), 1= 4 times, 2=8 times,3=16times,4=32 times
Range setting	5	Unsigned integer	1	Read/write	03/06	0=Auto(default),1=20uS/cm, 2=200uS/cm,3=2000uS/cm, 4=10000uS/cm
Temp. mode	6	Unsigned integer	1	Read/write	03/06	0=Automatic, 1=Manual
Conductivity offset	7	floating point	2	Read/write	03/16	default is 0.0 uS/cm
Conductivity factor	9	floating point	2	Read/write	03/16	default is 1.0
Manual temp. value	11	floating point	2	Read/write	03/16	Default 25°C
Temp. offset	13	floating point	2	Read/write	03/16	default is 0.0°C
Sensor constant	15	floating point	2	Read/write	03/16	Default value 1.0
TDS conversion factor	17	floating point	2	Read/write	03/16	Default value 0.5
Temp. Comp. factor	19	floating point	2	Read/write	03/16	Default value 1.85%/°C
Zero point cal	21	floating point	2	Write-only	03/16	Write 0 to perform zero CAL
The 1 st point cal	23	floating point	2	Read/write	03/16	Write 84uS/cm to start CAL
The 2nd point cal	25	floating point	2	Read/write	03/16	Write 146uS/cm to start CAL
The 3rd point cal	27	floating point	2	Read/write	03/16	Write 1413uS/cm to start CAL
The 4th point cal	29	floating point	2	Read/write	03/16	Write 4239uS/cm to start CAL
The 5th point cal	31	floating point	2	Read/write	03/16	Write 12880uS/cm to start CAL
Reset CAL	33	Unsigned integer	1	Write-only	06	Write 33 to start recovery calibration

Note 1: When the parity bit is set to "None", the stop bit can be set to 1STOP/2STOP;

When the parity bit is set to "Odd" or "Even", the stop bit can only be set to "1STOP".



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